

Tenure hero pipeline — MBB mirror (use existing panel first)

Last synced: 2026-09-02

Audience: Charles + COMPASS (Mac analysis).

Standalone: Step-by-step map from rsync'd `faculty_panel_with_pools.jsonl` to tenure HERO plots, mirroring MBB post-re-entry work without re-scraping.

MBB analogue doc: `../MBB_empirical_roster_select_replay.md`

Side-by-side ladder

Rung	MBB (done or in progress)	Tenure (this campaign)
Data spine	MBB panel CSV / exports	<code>faculty_panel_with_pools.jsonl</code>
Pass A HERO	<code>pass_a_empirical_bundle.py</code> → <code>sports_sandbox/hero/</code>	<code>tenure_pass_a_hero.py</code> → <code>tenure_sandbox/hero/</code>
Pass A F-HERO	<code>pass_a_congestion_conditional.py</code> → <code>fhero/</code>	Act II — T_j at fixed $\hat{A}$ (pending lock)
ρ / H_sort	Reigning hero calibration; $\rho^* \approx 0 \rightarrow$ park for MBB	Act III — run diagnostic; park only if flat
Generative replay	<code>reigning_hero_sim_hero.py</code>	Act IV — after Alex lock
Scrape / expand	Box QC, min20, mg10	Parallel — PEER on Rivanna when scheduled

PD29 delta (Sep 2, 2026) — Alex locks vs v0

Source: `transcripts/PD29_notes.md` — Charles showed **dirty porch** same call; Alex approved assortment direction, then tightened tenure mechanics.

Dimension	v0 (this doc, Sep 1)	PD29 lock (exploratory → next build)
Unit of analysis (HERO row)	One row per person (ASST-PS mean peer LOO) or last assistant year on some runs	One row per decision event (~assistant year 5–6 with tenure or out signal)
Own $\hat{A}$	ASST-PS mean pubs/year over assistant years; LAST-PS cum on some exploratory HERO runs	ALL-PS · mean · own pubs at decision year = <code>pubs_per_career_year</code> (cum ÷ publishing-years since first pub)
Pond / peer pool	Unixyear assistant peer pools; LOO over assistant person-years	Whole department in decision year (all faculty ranks for peer norm); not every historical deptxyear
Outcome frame	Option A: censored out of rates , in quantile rank ; scrape-window censor	Up-or-out in decision year (army-like); cumulative productivity “until that point”
Data spine	<code>faculty_panel_with_pools.jsonl</code> person-year	Master authorxyear pubs table → precompute running cum / rates; slice for decision years
Porch (Sep 2)	BDP + HERO deck on existing panel	Alex: overlap H_sort high ✓; censored = missing scrape; promoted-with-zero annual rate = wrong metric → supports T3

COMPASS sequencing: Keep v0 porch/HERO as **Act I exploratory** (Alex saw it Sep 2). **Do not** treat v0 LAST-PS / ASST-PS bars as paper-final without PD29 spine or explicit sensitivity. Next engineering tranche = **T5 master table** + **T1–T4 decision-year slice**.

Open (PD29, not locked): Homophily on all faculty vs assistant-only pond; exact early/late window around year 6; survival analysis deferred.

Grain naming (Charles lock — Sep 2)

Aligned with MBB **ALL-PS** / **LAST-PS** vocabulary; tenure adds **ASST-PS** because assistant-only person-years are the main v0 panel.

Windows (person-years)

Token	Label	Meaning
<code>all_ps</code>	ALL-PS	All person-years in panel (any rank)
<code>asst_ps</code>	ASST-PS	Assistant-rank person-years only
<code>last_ps</code>	LAST-PS	Final assistant person-year (exit cross-section)

Stats

Token	Label	Meaning
mean	mean	Average over the window (person collapse for ASST-PS)
cum	cum	Stock through end of window
annum	annum	Single person-year value (replaces legacy "annual")

Retired everywhere: spell-mean, spell_mean, grain=... in favor of --window / --stat .

Canonical combinations (v0 + PD29)

Use	Window · stat · metric	Code / field
v0 HERO peer X	ASST-PS · mean · peer LOO (annum)	--window asst_ps --stat annum (default)
LAST-PS HERO peer X	LAST-PS · cum · peer LOO	--window last_ps --stat cum
LAST-PS ability slice	LAST-PS · cum · own pubs	--window last_ps --x-metric own_cum
Alex Â (PD29)	ALL-PS · mean · own pubs at decision year	pubs_per_career_year in author_year_career_master.jsonl

Decision year = cohort filter / which calendar year we read the metric — not a separate window token.

Code: tenure/scripts/tenure_grain_labels.py · CLI tenure/scripts/tenure_pass_a_hero.py --window / --stat

Step 0 — Confirm Mac has panel (once per session)

```
./scripts/rsync_pull_recent_hpc.sh tenure # if unsure
ls -lh tenure/tenure_pipeline/faculty_panel_with_pools.jsonl
```

See 20260901_PEER_to_COMPASS_tenure_hero_mac_handoff.md .

Step 1 — Inference filter (strict v0)

From person-year rows, keep rows where:

- match_confidence ∈ {HIGH, MEDIUM}
- poolq_loo_mean is not null (on assistant rows used for LOO)
- ever_assistant is true

Do not hero-plot raw 106K without understanding OpenAlex NONE tier (62K person-years).

Step 2 — Collapse to person-level (HERO window)

Same logic as tenure/tenure_pipeline/stage9_analysis.py :

- One record per faculty_id
- loo_mean = mean of poolq_loo_mean over assistant years with non-null LOO (ASST-PS · mean · peer annum)
- Drop persons with no computable LOO assistant years
- Person flags: tenure_event, attrition, censored

Step 3 — Outcomes (Option A — locked for v0)

Resolved = tenure OR attrition (known ending).

Rate	Numerator	Denominator
Tenure rate	tenure_event	resolved in bin
Attrition rate	attrition	resolved in bin

Censored: report n_censored per bin; exclude from rate denominator. Survival / Cox later.

Two separate decisions (do not conflate)

Decision	v0 lock	What it controls
A — Rate denominator (Option A)	Censored out of tenure/attrition rate	Bar height = tenured ÷ (tenured + attrition) in bin
B — Quantile assignment	Censored in when ranking on LOO	Who lands in ventile 1...K (n_all ≈ equal); slide label resolved/all

Option A is about Y (outcome unknown → no rate). Decision B is about X (peer context known → still get a bin).

Censored in quantile bins — pros/cons for Alex

Question for Alex: Should **right-censored** faculty (still assistant near end of scrape window, ~2022–2024 last-asst years; no observed tenure or attrition yet) **stay in the pool used to define quantile cutpoints**, while remaining **excluded from the rate denominator** (current v0)?

What censored means here (not survival analysis yet):

- **Tenured** — promoted associate/full within `gap_tolerance` (2 yr) of last assistant year
- **Attrition** — last assistant year **before** window tail, never promoted
- **Censored** — never promoted **and** still assistant near `max_year` (2024) → outcome not yet observed

On the last-ps cum LOO panel (N=732): **408 censored (56%), 324 resolved**. Censorship is **not** uniform across LOO (middle ventiles can be `4/37` resolved vs `26/37` elsewhere).

Pros — keep censored in quantile assignment (current)

1. **X reflects the full inference panel.** LOO is defined for censored people; their peer context is real. Dropping them before binning changes who defines “ventile 10.”
2. **Avoids selection bias in cutpoints.** Censored correlate with LOO and career stage (recent hires still in the pipeline). Excluding them before ranking shifts ventile boundaries — especially if high-LOO or mid-LOO assistants are disproportionately still active.
3. **Parallel to MBB HERO.** Everyone with computable roster X gets a ventile; draft Y is only measured where the outcome exists. Same split: known X, partial Y.
4. **Equal `n_all` is interpretable.** “This ventile is the bottom 5% of the **sample by LOO**” includes people still on the job market clock — which matches “who faces this peer environment in our data?”
5. **Transparent accounting.** CSV + slides report `n_censored`, `n_resolved`, and `resolved/all` per bin; compositional skew is visible rather than hidden by pre-filtering.

Cons — keep censored in quantile assignment

1. **Unequal resolved N despite equal `n_all`.** Quantile guarantees ~equal people per bin, **not** equal resolved counts → noisy Wilson CIs where censorship piles up (e.g. bin 11: `4/37`).
2. **Harder slide read without training.** Looks like “broken quantile” if you only see resolved counts (fixed in slides with `resolved/all` + footnote).
3. **Compositional confound in rates.** A low tenure rate might mix “true lower promotion among leavers” with “this ventile is mostly young / still assistant” — not separable in binned Wilson plots alone.
4. **Not the only defensible choice.** Alternatives are easy to run as sensitivity (see below).

Main alternatives (sensitivity, not v0)

Variant	Bin assignment	Rate denominator	Tradeoff
v0 (current)	All with LOO	Resolved only	Full X panel; variable resolved n
Resolved-only quantile	Re-rank LOO on resolved only	Resolved	~Equal resolved per bin; cutpoints ignore censored LOO → shifted ventiles
Resolved-only panel	Quantile on LOO	Resolved	Drops ~56% (last-ps); different population (“careers already ended”)
Option B rates	All with LOO (or any)	All incl. censored	Treat censored as non-tenure → not recommended without survival frame

COMPASS view: v0 is coherent for **exploratory HERO** (shape of tenure vs peer context among cases with known endings, indexed to the full LOO spectrum). For **paper claims**, Alex should sign off on (i) whether resolved-only quantile sensitivity is required, and (ii) whether last-ps HERO needs a **cohort aperture** that reduces censorship (e.g. last-asst year cap) before debating bin mechanics.

Code: `stage9_analysis._bin_rows` ranks **all** persons with LOO; `_aggregate_bins` applies Option A rates. Flags from `panel_builder.py` (`censored` if `last_asst >= max_year - gap_tolerance` and no tenure event).

Step 4 — Bin and plot

Knob	v0
X	Person-level <code>loo_mean</code>
Method	Quantile — equal <code>n_all</code> per bin (censored included in rank; see § censored pros/cons)
Count	16
CI	Wilson on resolved counts
Outputs	PNG + CSV + provenance JSON → <code>tenure_sandbox/hero/</code>

Stage 9 (`stage9_inverted_u.png` , Apr 2026) used 18 bins on a **broader** population — useful sanity check, **not** the v0 deliverable.

Step 5 — Act II+ (later)

Act	Action
II F-HERO	Bin department mean pubs at fixed $\hat{A}$ slice
III ρ	Department-level H_{sort} bracket (compare to MBB reigning hero ρ folder)
IV Sim	Frozen departments → fit score knobs → sim tenure flags → replay HERO

ρ policy: Do **not** assume $\rho \approx 0$ on tenure. Run Act III before parking ASSIGN like MBB.

Code map

File	Role
<code>tenure/tenure_pipeline/stage9_analysis.py</code>	Reference implementation (collapse + bin + plot)
<code>tenure/tenure_pipeline/build_faculty_panel_inference_v1.py</code>	Inference tier filter reference
<code>tenure/scripts/tenure_grain_labels.py</code>	Window + stat labels (ALL-PS / ASST-PS / LAST-PS · mean / cum / annum)
<code>tenure/scripts/tenure_pass_a_hero.py</code>	v0 CLI → <code>tenure_sandbox/hero/</code>
<code>sports/scripts/pass_a_empirical_bundle.py</code>	MBB template for provenance + filenames

Thread log

Date	Entry
2026-09-01	Initial pipeline map; v0 locks from Charles + Alex direction.
2026-09-01	§ censored in quantile bins — pros/cons for Alex (Option A rates + include censored in rank).
2026-09-02	§ Grain naming lock — ASST-PS / LAST-PS / ALL-PS + mean / cum / annum; retire spell-mean; fix Alex $\hat{A}$ = ALL-PS mean at decision year.